The Syllabus to Quiz Generator (Multi Agent System)

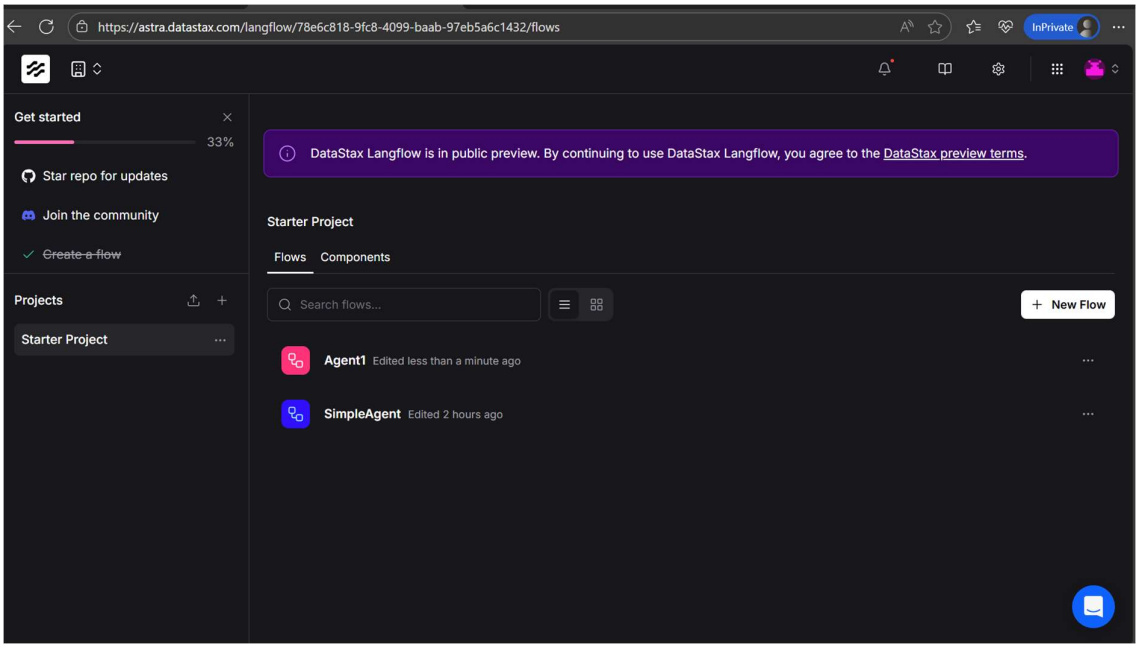
Project on Education domain: "The Syllabus to Quiz Generator" (Multi-Agent solution)
Problem statement - Teachers or students often have a broad topic (e.g., "Photosynthesis") but need to break it down into a structured lesson plan and then immediately generate assessment questions for it. Doing this in one prompt often results in lower quality for both.
Solution - Agent 1 (The Architect): breaks the topic into a structured outline.
Agent 2 (The Examiner): takes that outline and creates a specific quiz based on it.

Prerequisites:

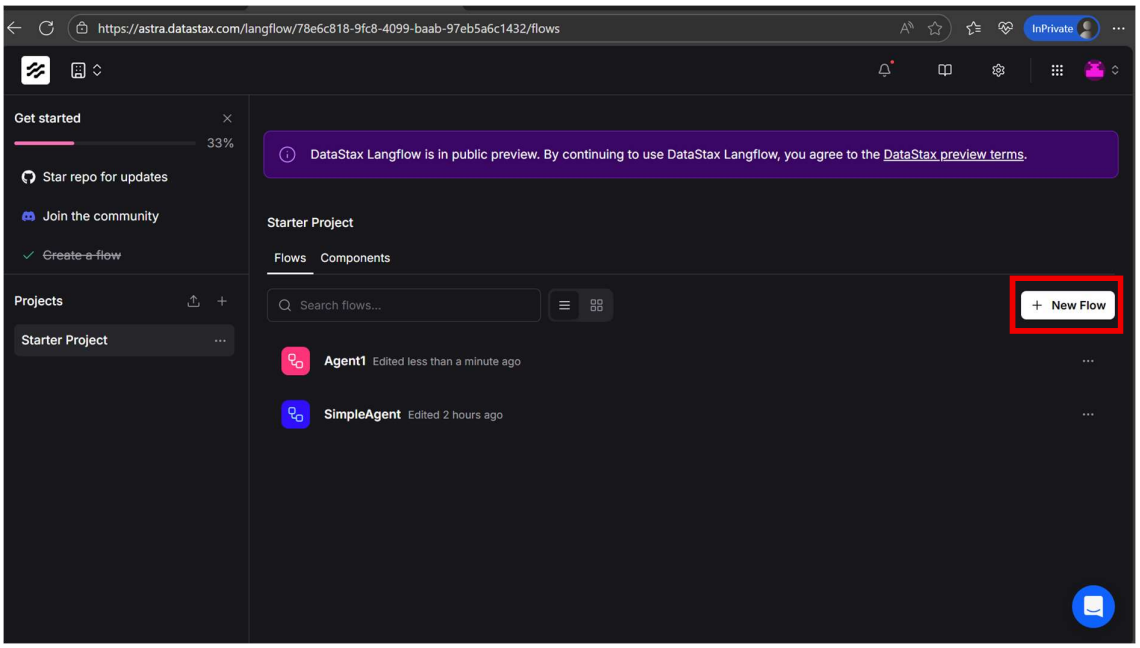
- Watsonx API Key from IBM Cloud
- Watsonx Project ID from IBM Cloud
- Watsonx API Endpoint for Dallas - https://us-south.ml.cloud.ibm.com

Steps to complete the project:

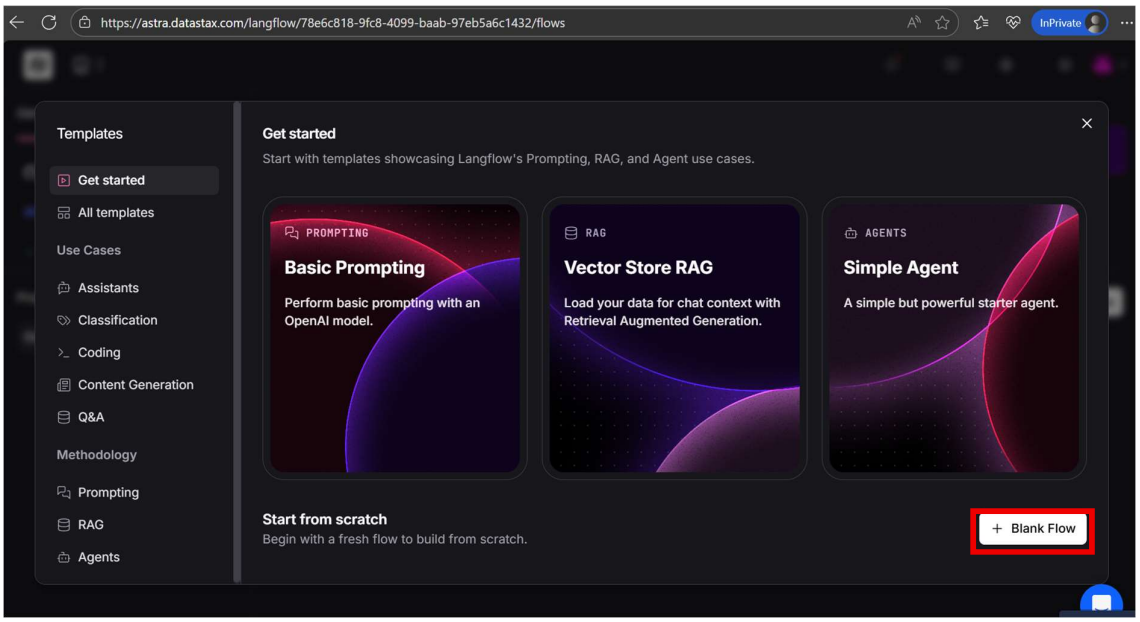
- Log in to Langflow



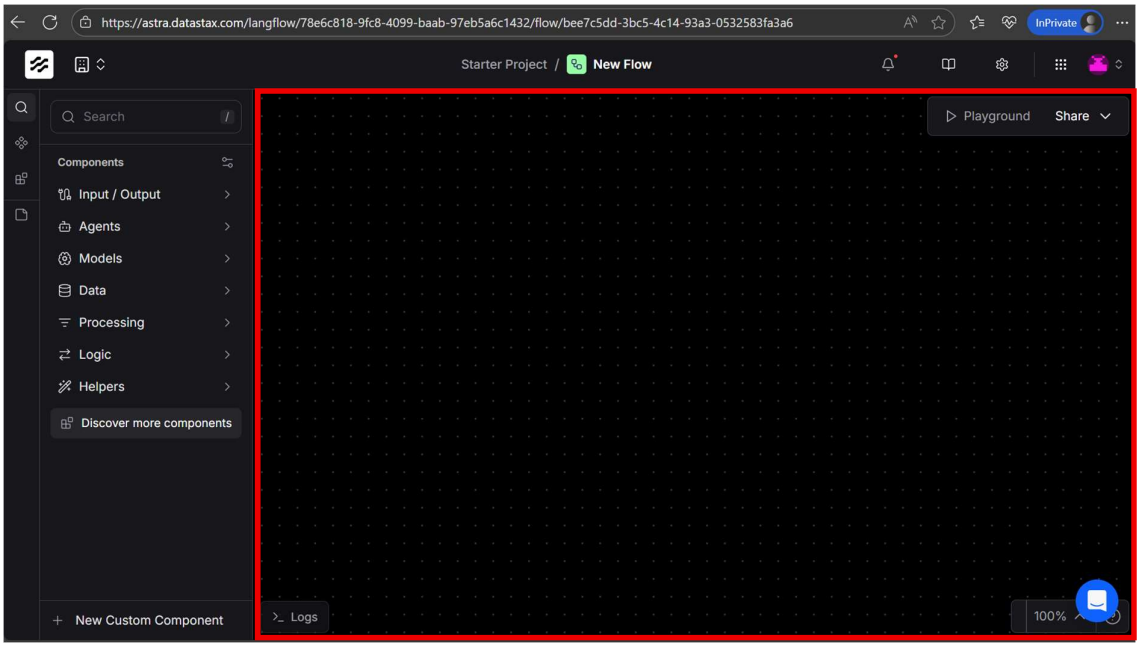
- Click on New Flow.



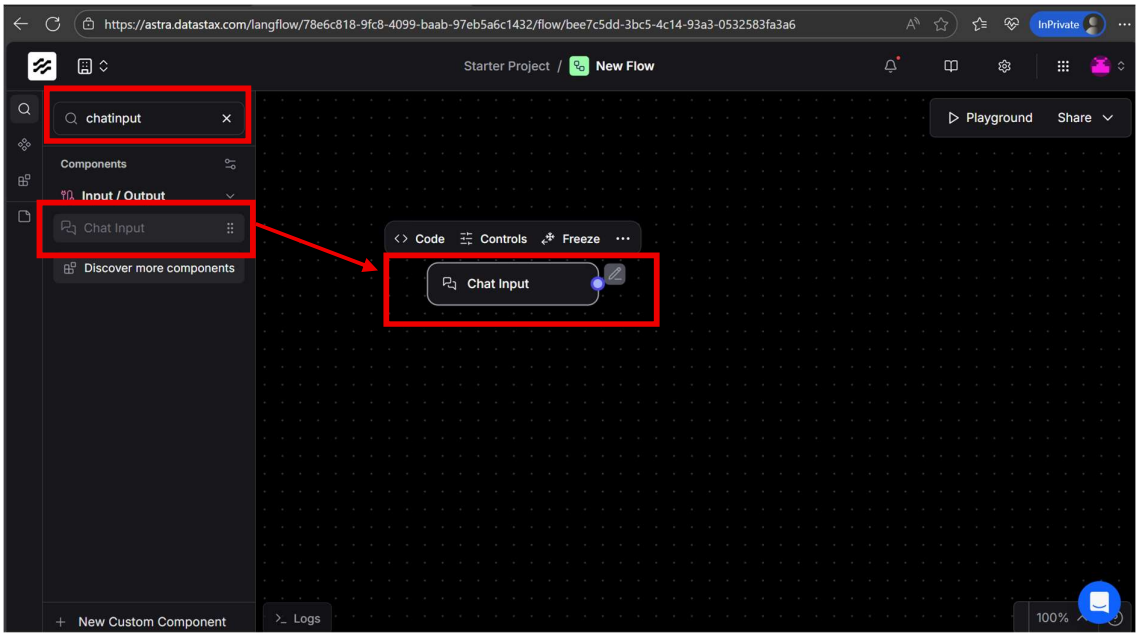
- Click on Blank flow.



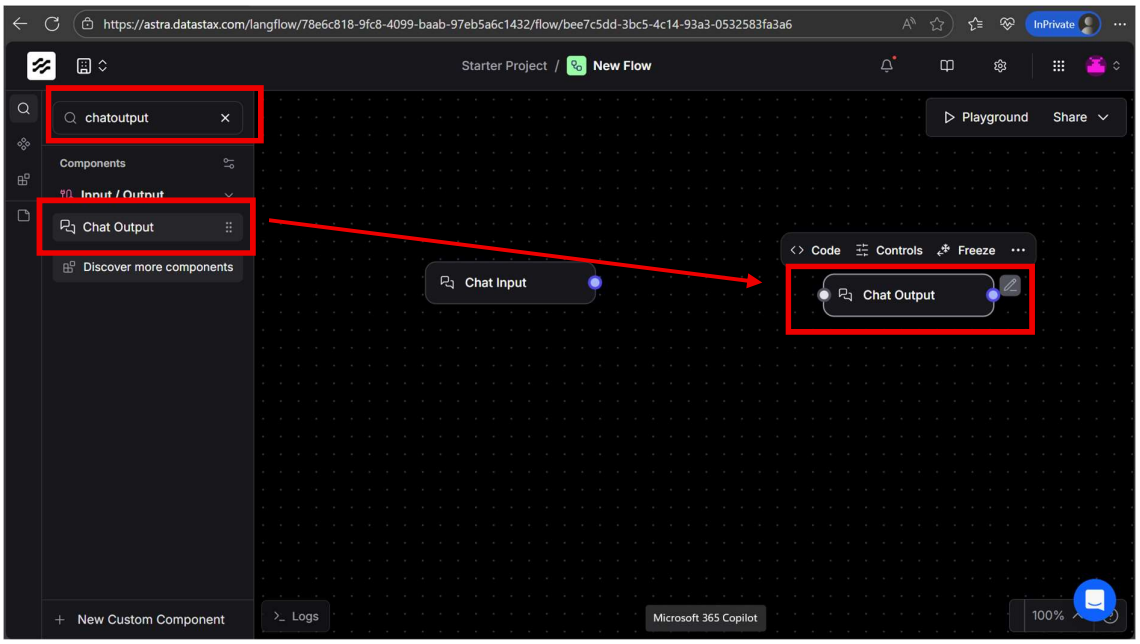
- Blank Canvas will be opened as follows.



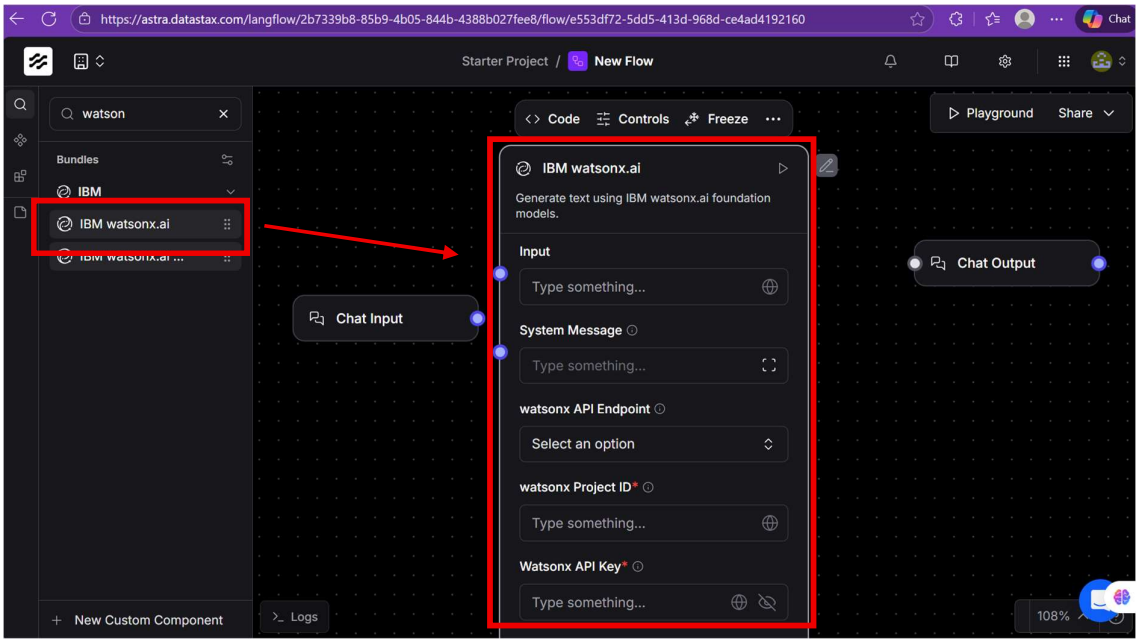
- Type **chatinput** in the search box which is there on the left navigation menu. Select and drag chat Input component to the Canvas. Paste it on the canvas.



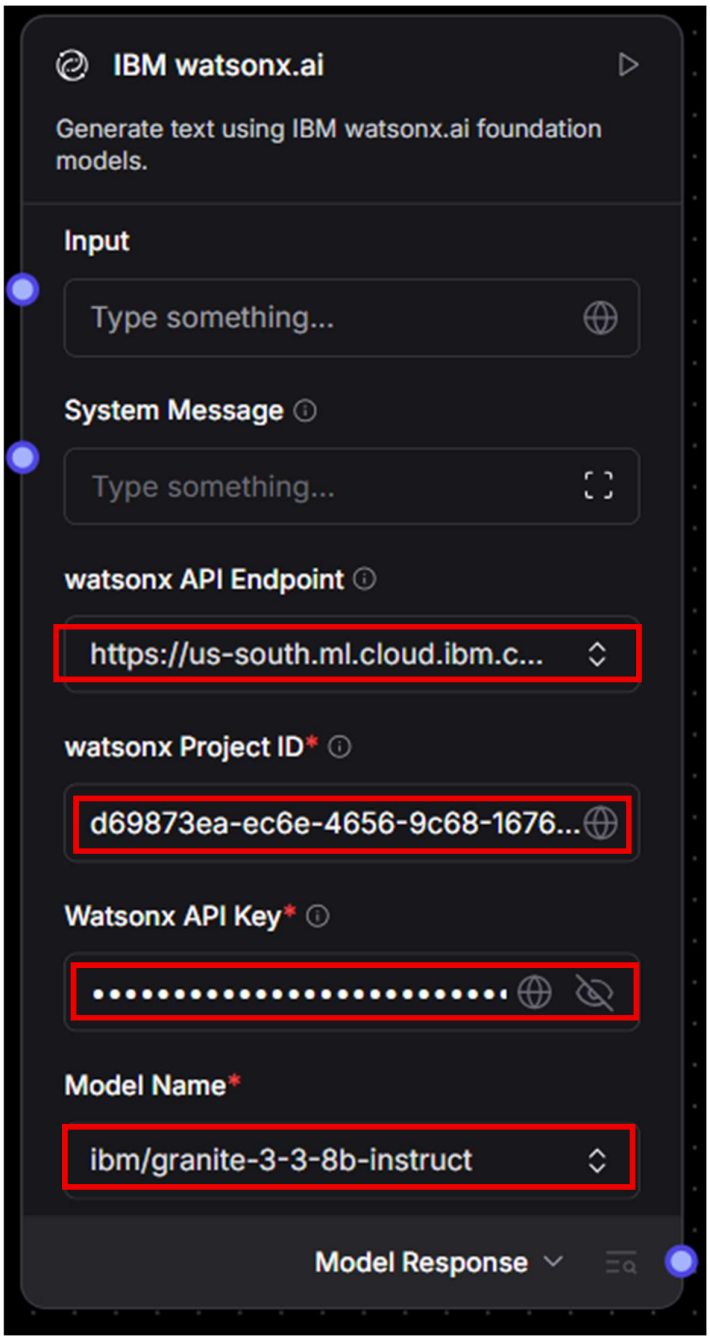
- Type **chatoutput** in the search box which is there on the left navigation menu. Select and drag chat Output component to the Canvas. Paste it on the canvas.



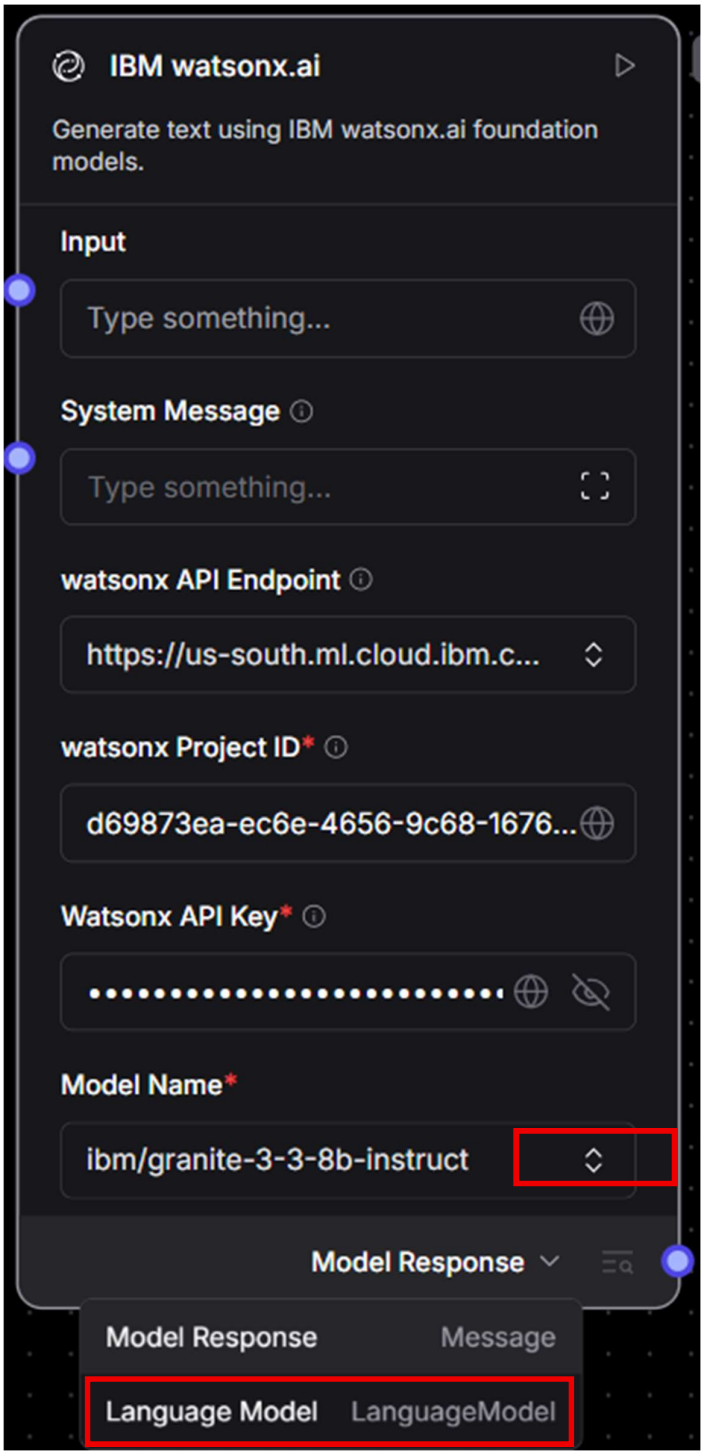
- Type **watsonx** in the search box which is there on the left navigation menu. Select and drag watsonx.ai component to the Canvas. Paste it on the canvas.



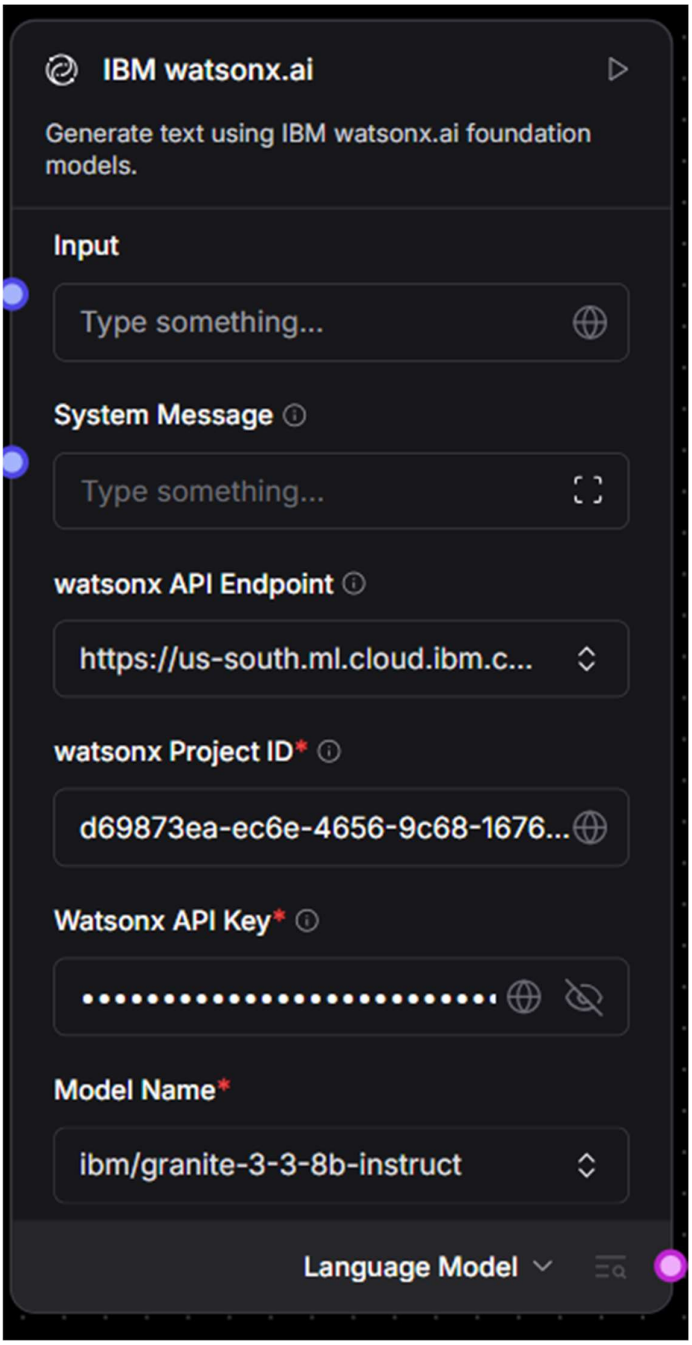
- Configure the IBM Watsonx.ai component with API Endpoint, Project ID, API Key and choose Model Name as **ibm/granite-3-3-8b-instruct**



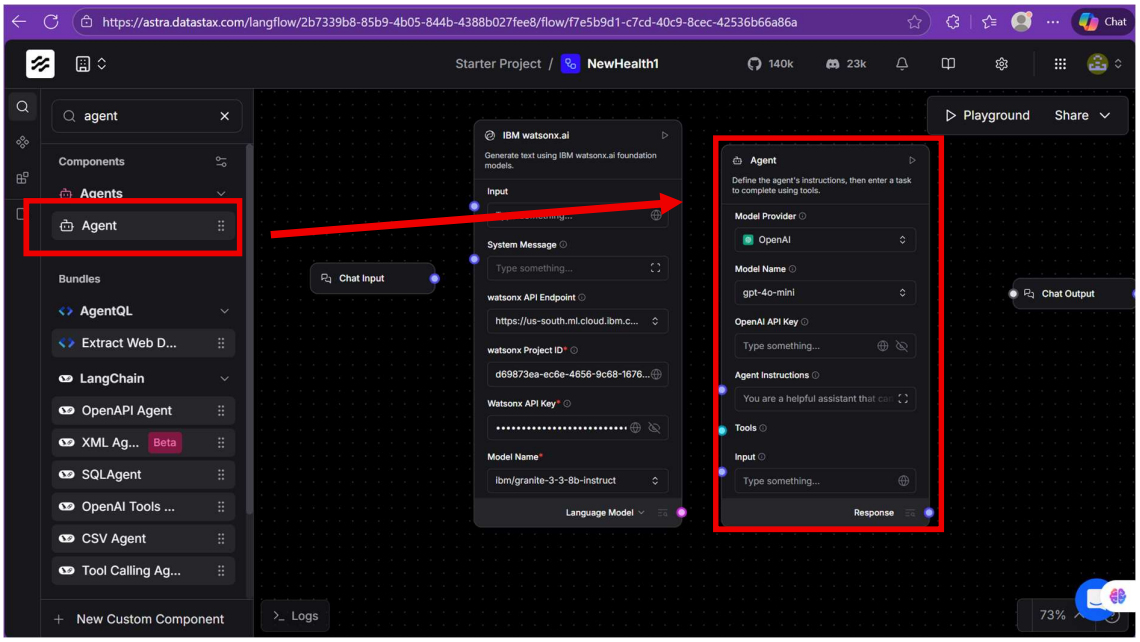
- Click on the arrow next to Model response of IBM Watsonx.ai component → change the Model Response to Language Model



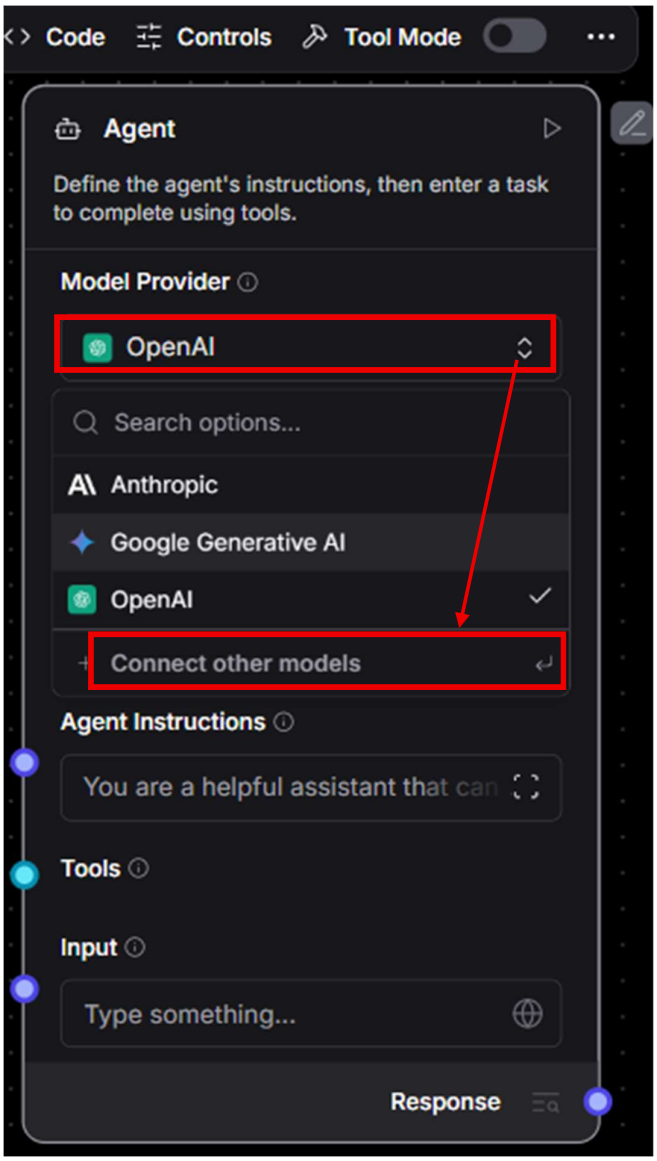
- Now IBM Watsonx.ai component looks as follows



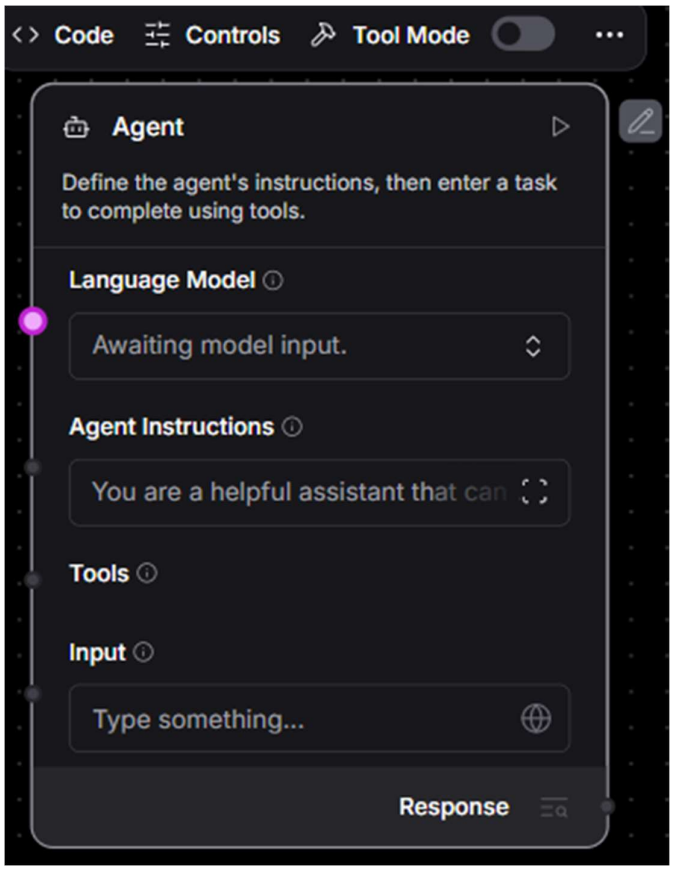
- Type **agent** on search box. Drag the agent on to the canvas and drop on the canvas. This is Agent1 for the project.



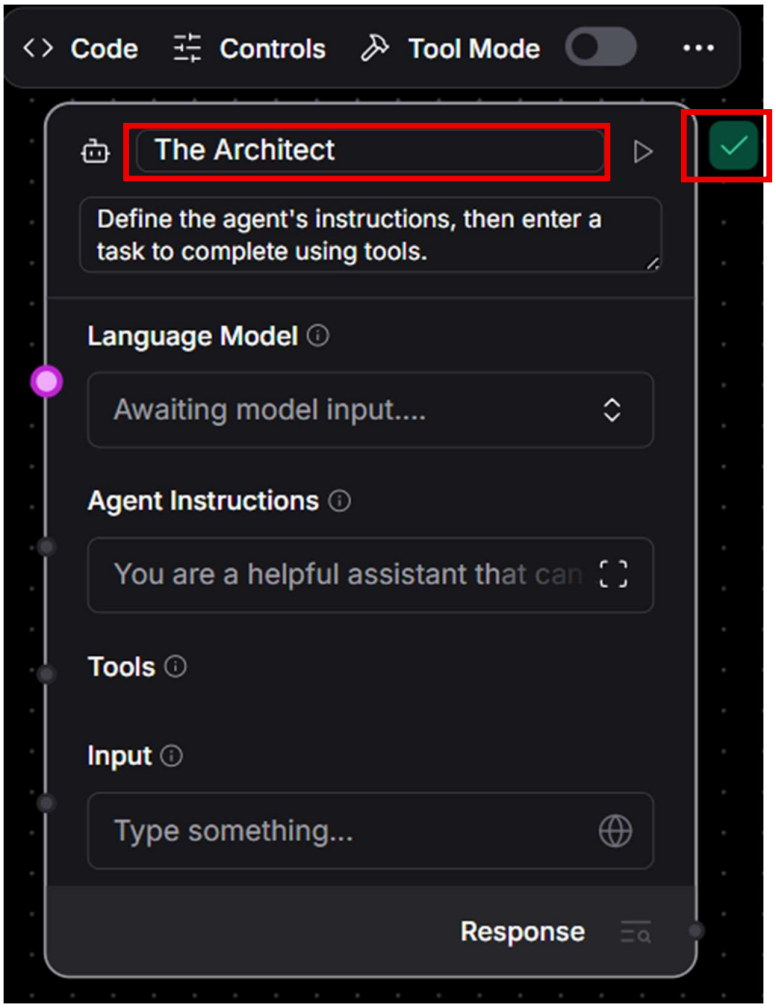
- Observe the Agent Component. The Model Provider is Open AI. But we need IBM Granite LLMs for the Project. To change for this, click on Open AI → select **Connect other models**



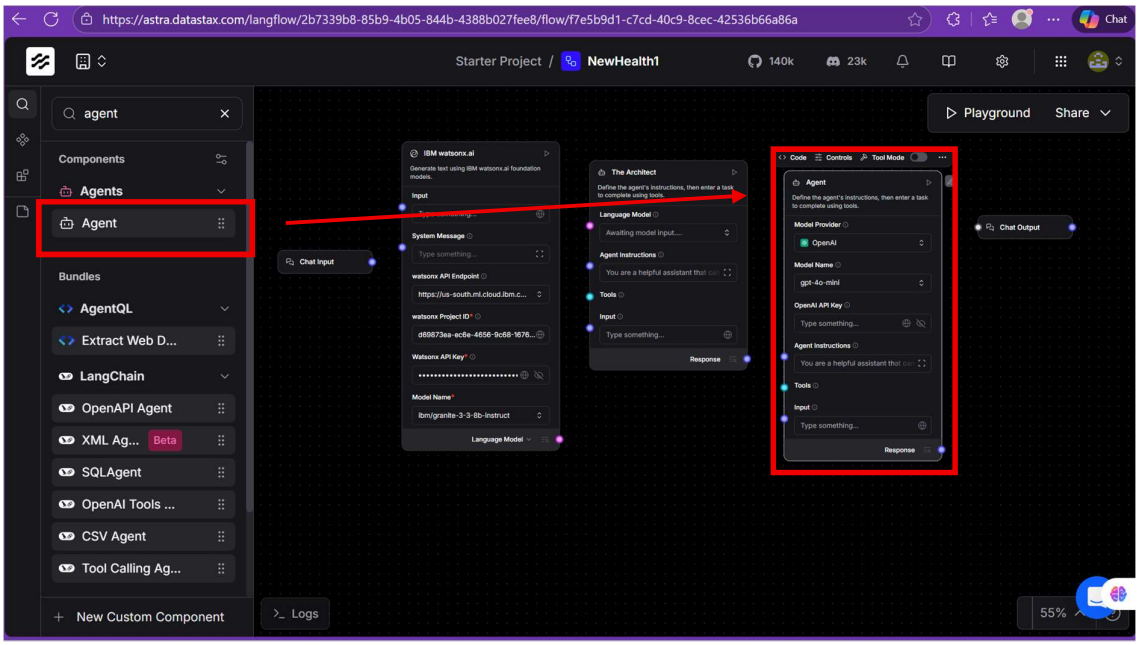
- Now the agent looks as follows



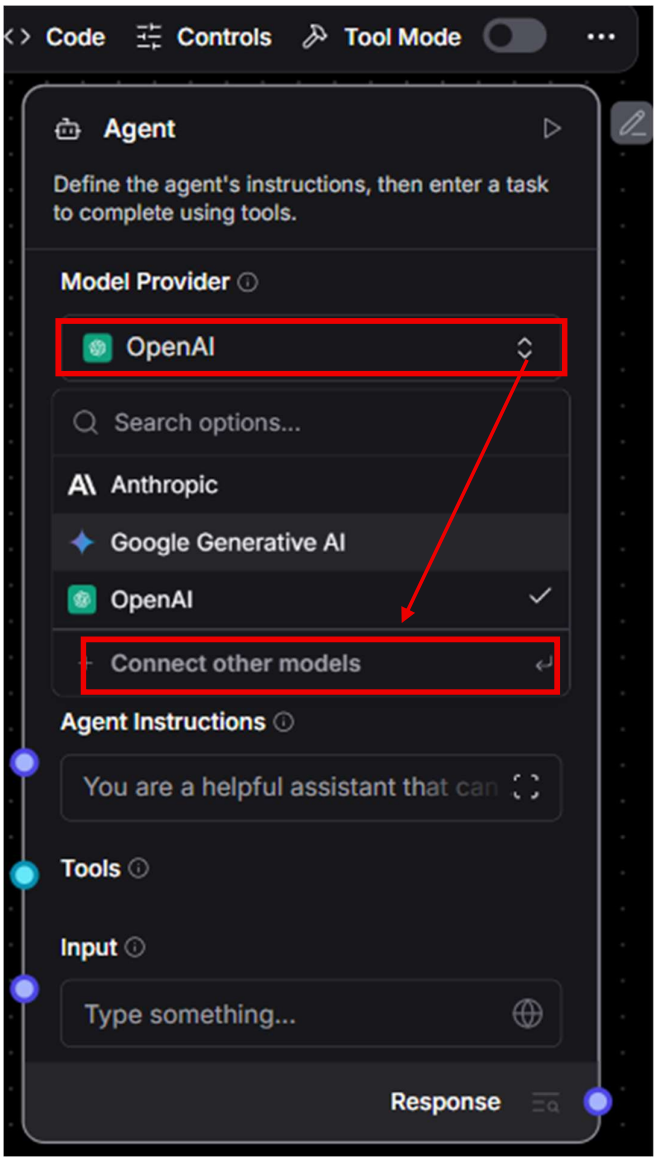
- Change the Name of the Agent to The Architect. To change the name, click on the edit symbol appearing at the top of the agent component → type the name of the agent as **The Architect** → Enter



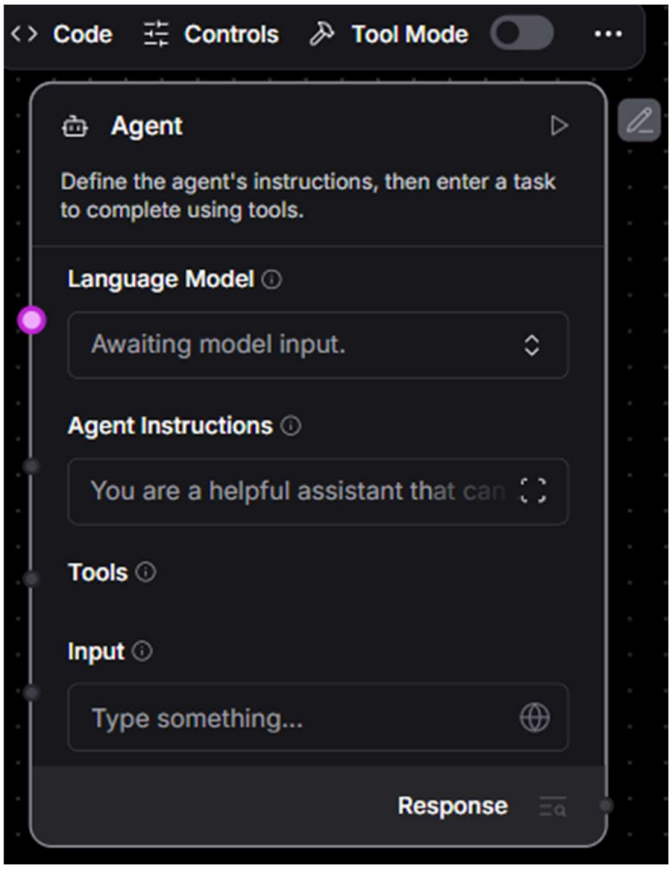
- Type **agent** on search box again. Drag the agent on to the canvas and drop on the canvas. This is Agent2 for the project.



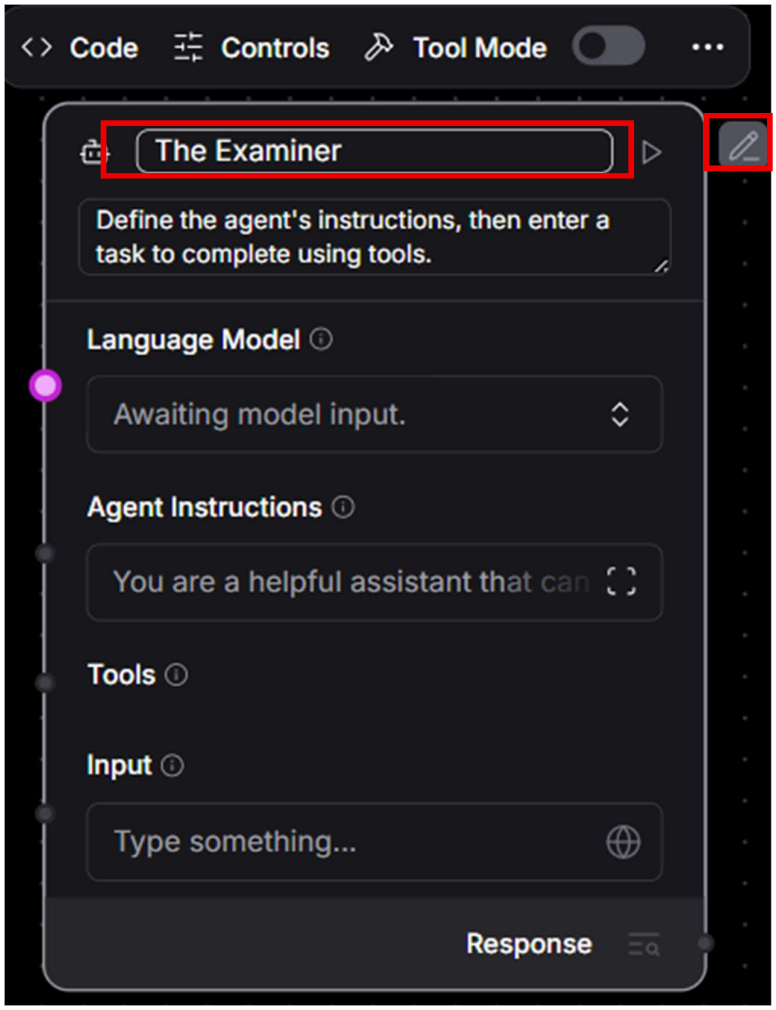
- Observe this Agent Component also. The Model Provider is Open AI. But we need IBM Granite LLMs for the Project. To change for this, click on Open AI → select **Connect other models**



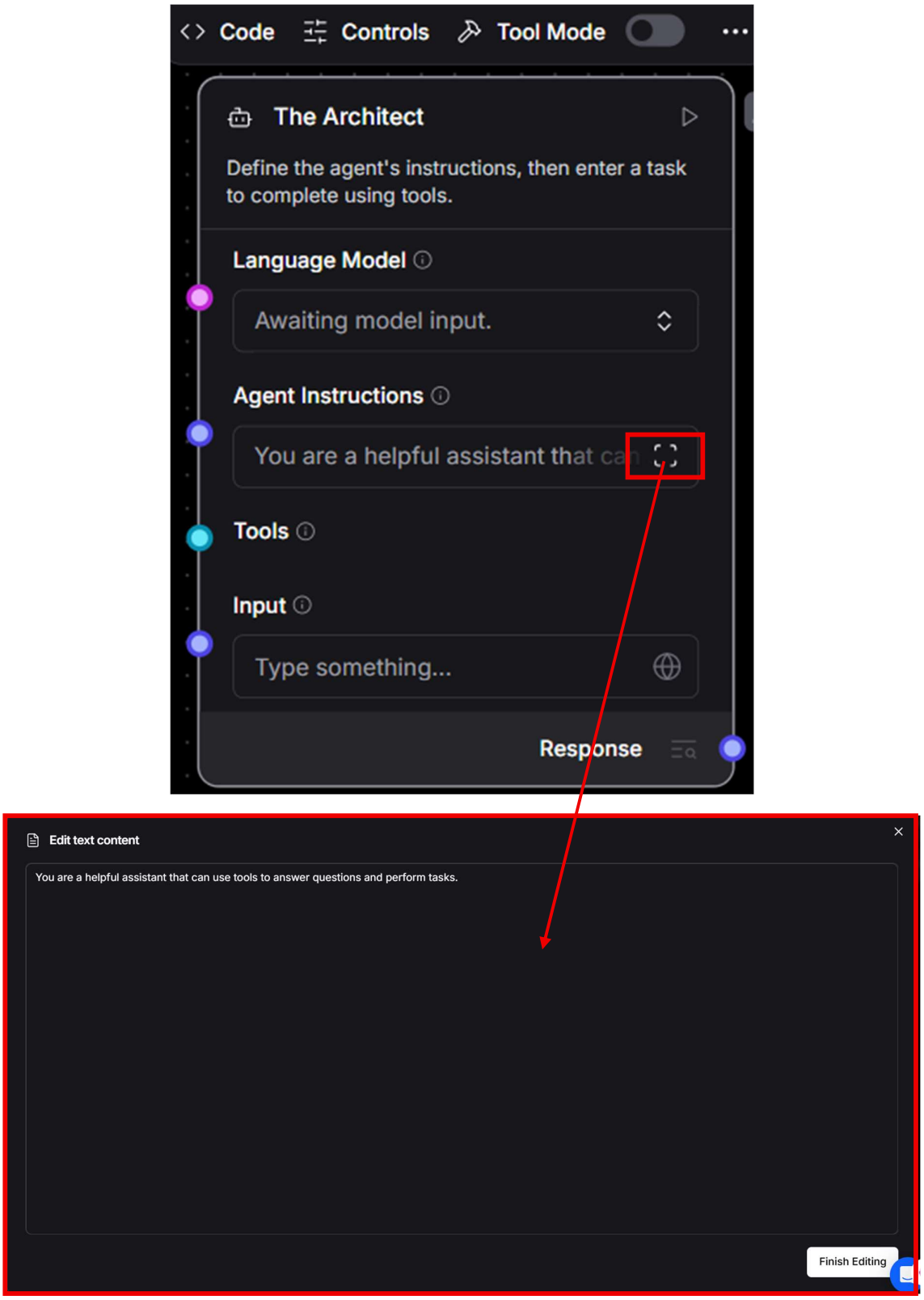
- Now the agent looks as follows



- Change the Name of the Agent to The Architect. To change the name, click on the edit symbol appearing at the top of the agent component → type the name of the agent as **The Examiner** → Enter



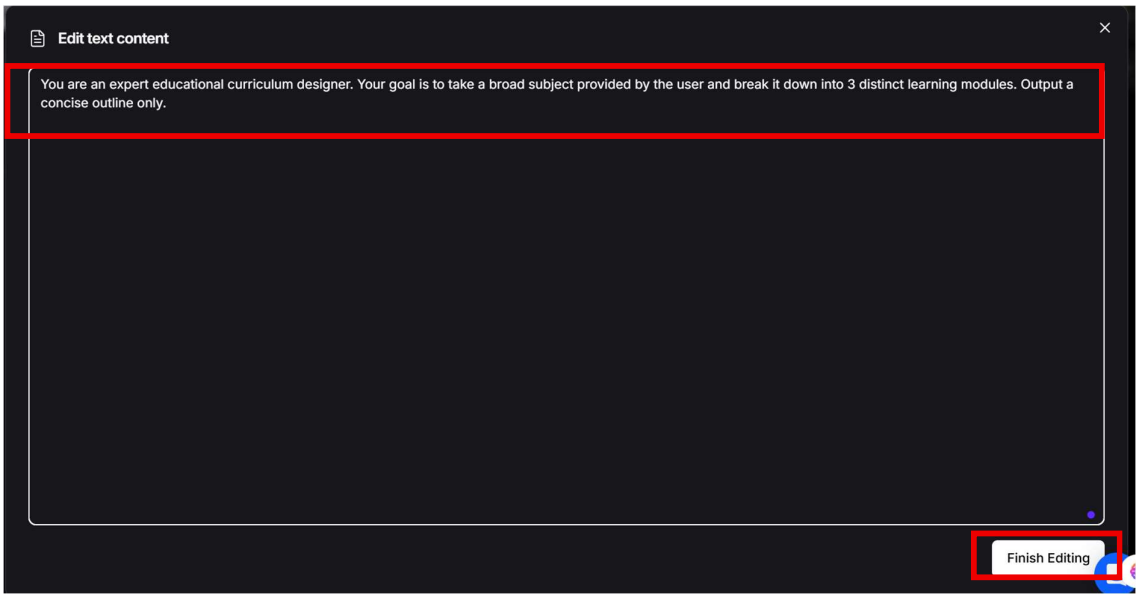
- Now select first agent (The Architect) → click on Square symbol below the Agent instructions → Edit text content window will appear



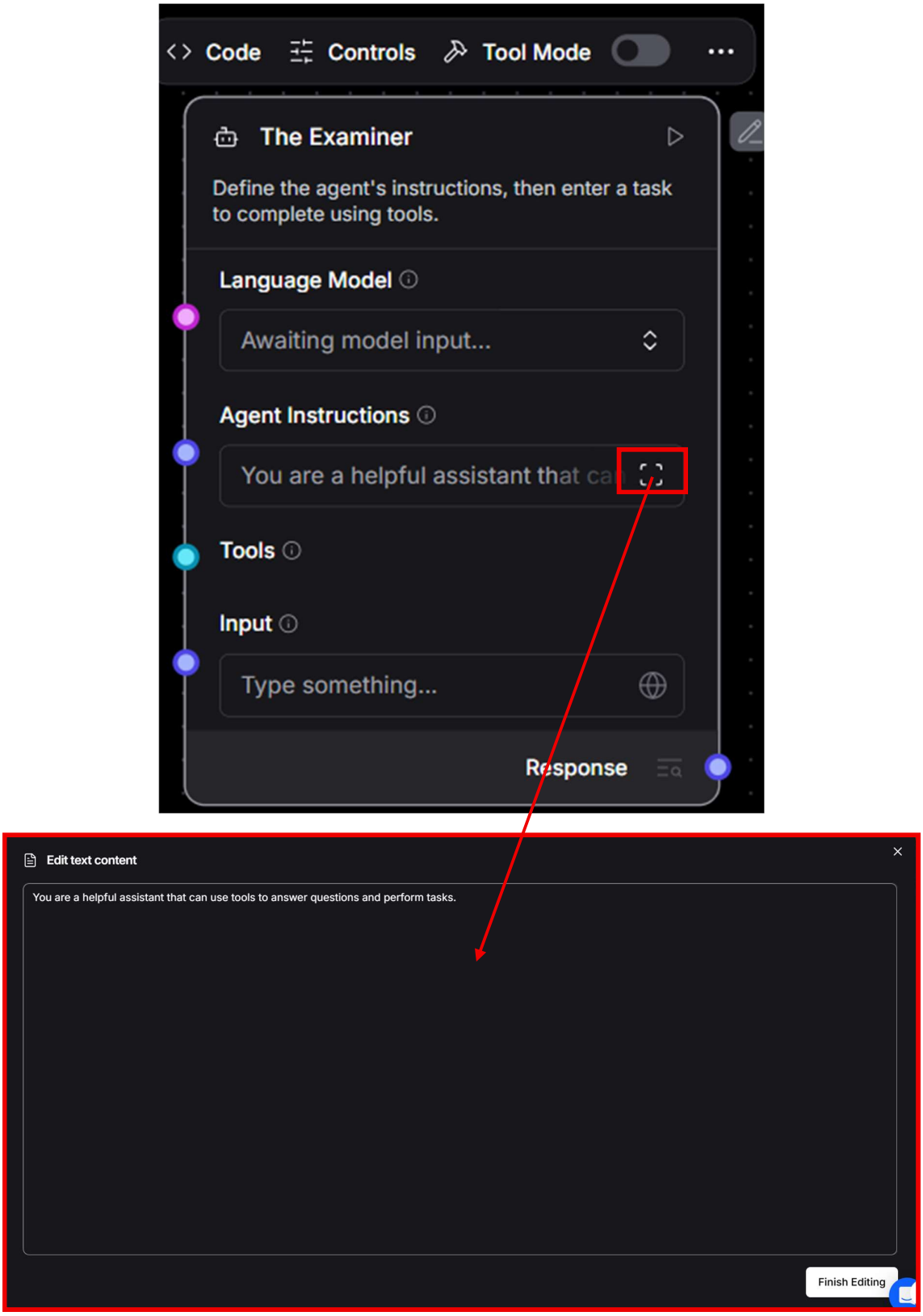
- Delete the default text → Enter the following text as instruction to the agent → click on Finish Editing button

Agent1(The Architect) instruction:

You are an expert educational curriculum designer. Your goal is to take a broad subject provided by the user and break it down into 3 distinct learning modules. Output a concise outline only.



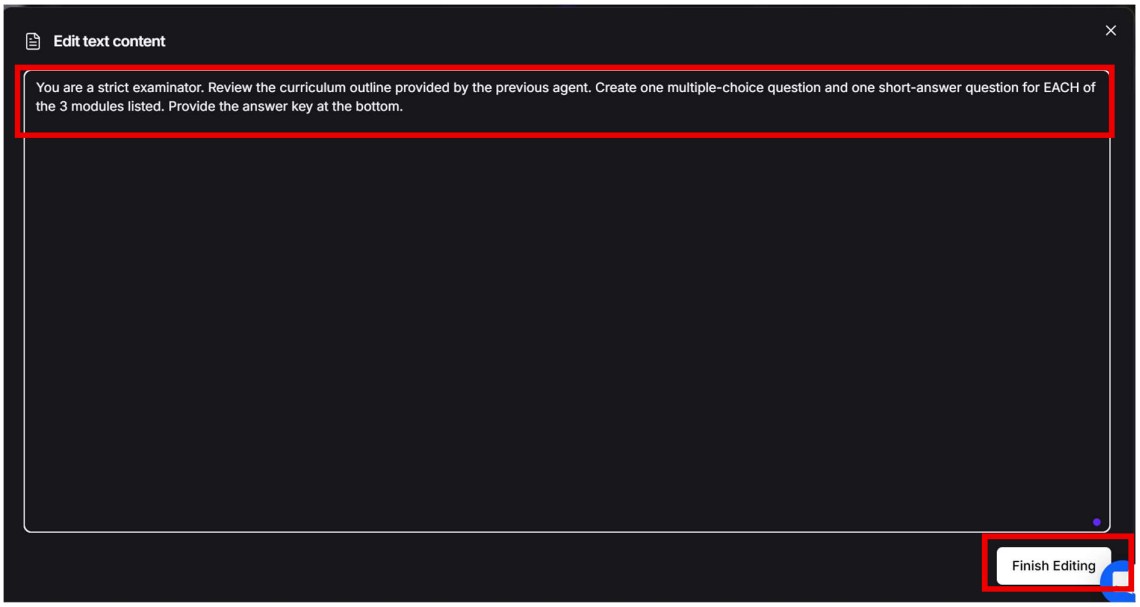
- Now select second agent (The Examiner) → click on Square symbol below the Agent instructions → Edit text content window will appear



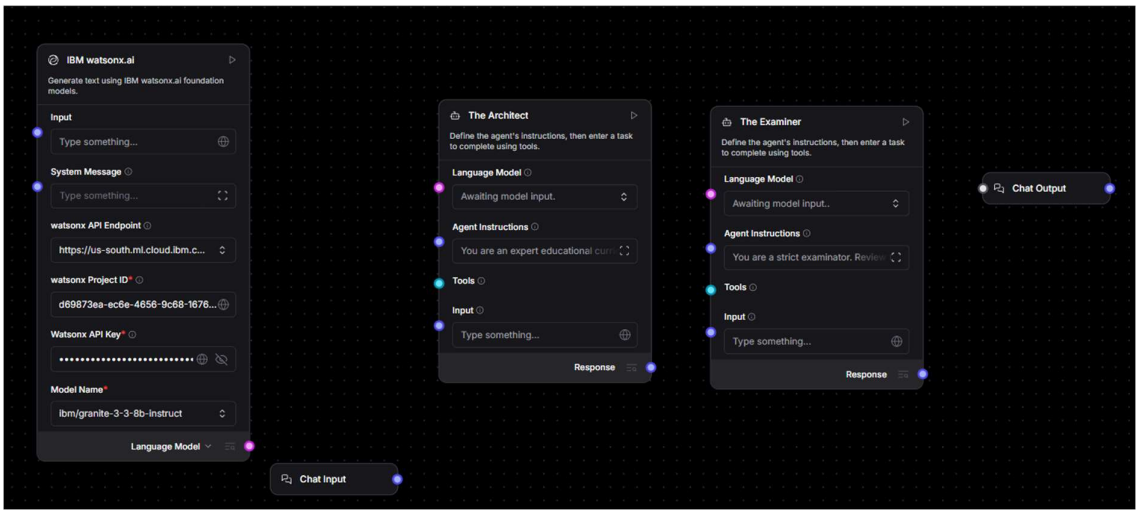
- Delete the default text → Enter the following text as instruction to the agent → click on Finish Editing button

Agent2(The Examiner) instruction:

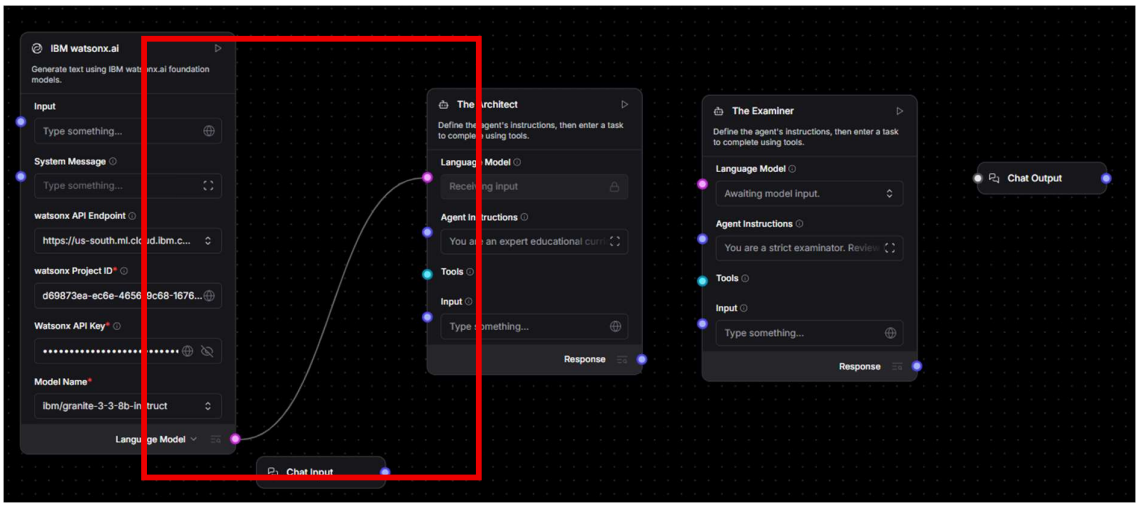
You are a strict examinator. Review the curriculum outline provided by the previous agent. Create one multiple-choice question and one short-answer question for EACH of the 3 modules listed. Provide the answer key at the bottom.



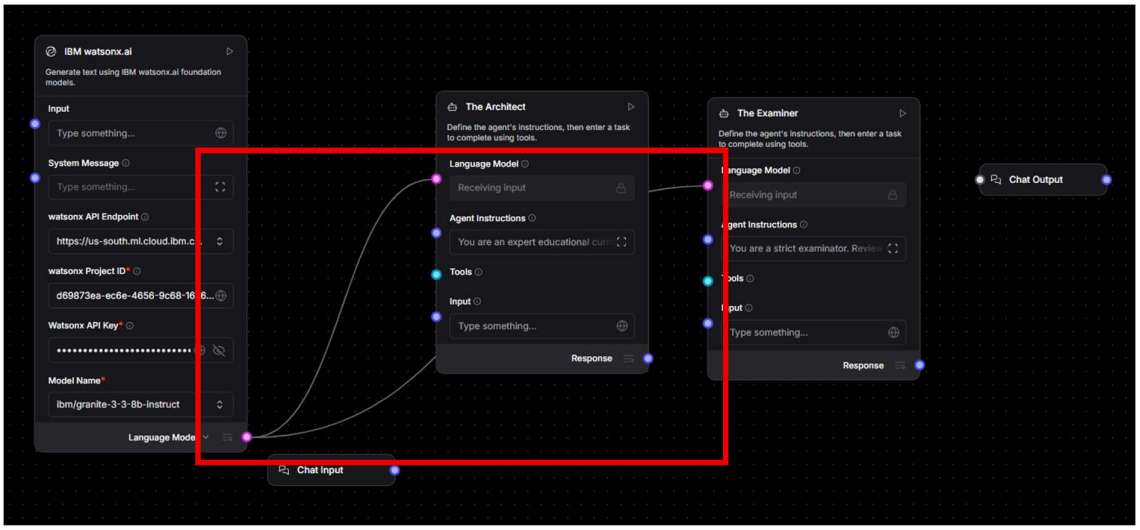
- Arrange the components on the canvas as below



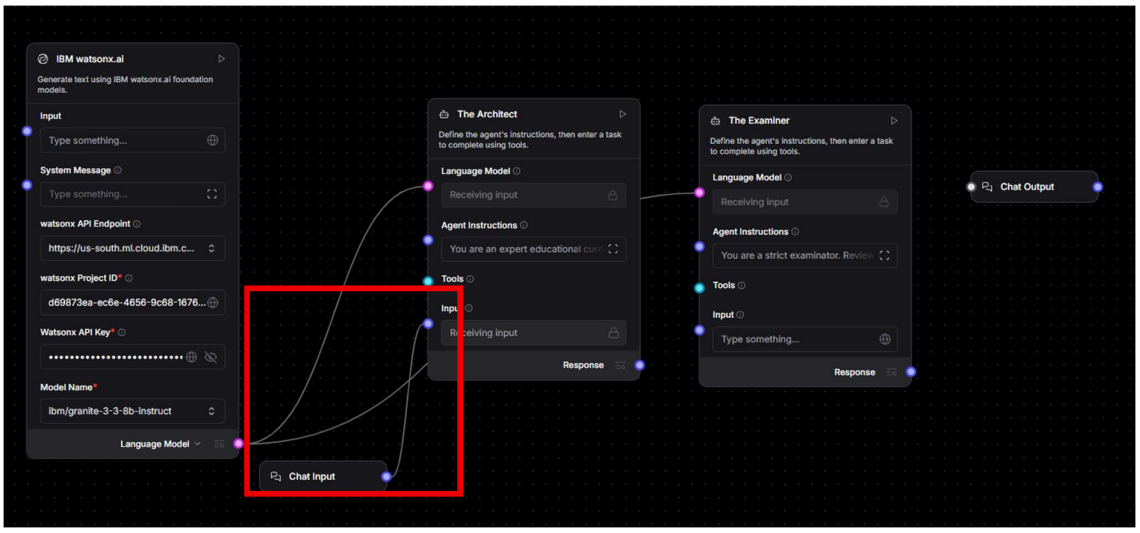
- Connect IBM Watsonx.ai component to agent1(The Architect)



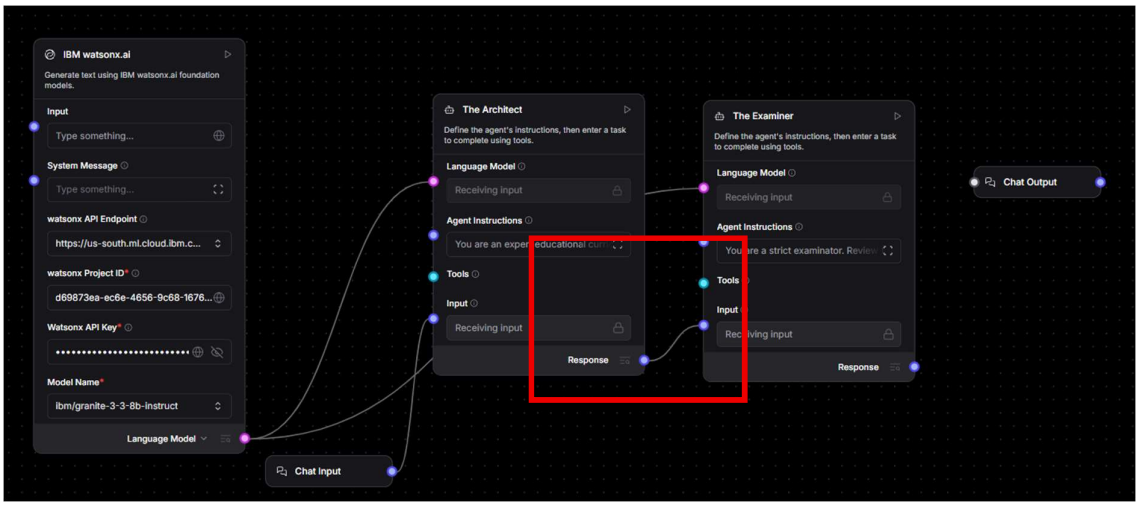
- Connect IBM Watsonx.ai component to agent2(The Examiner)



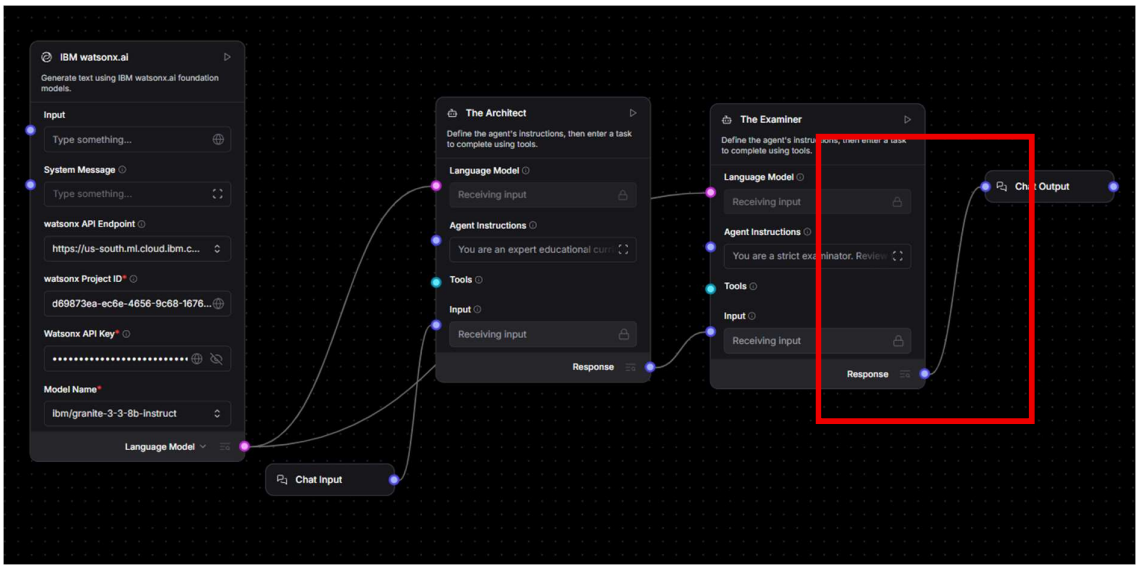
- Connect chat input to agent1(The Architect)



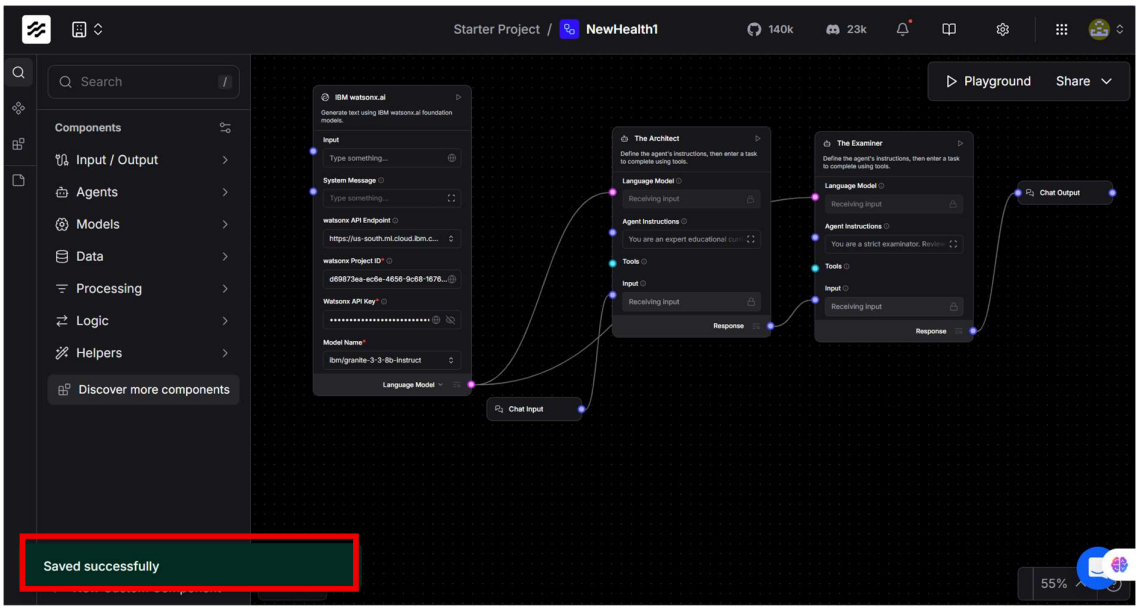
- Connect Agent1(The Architect) response to Agent2(The Examiner)



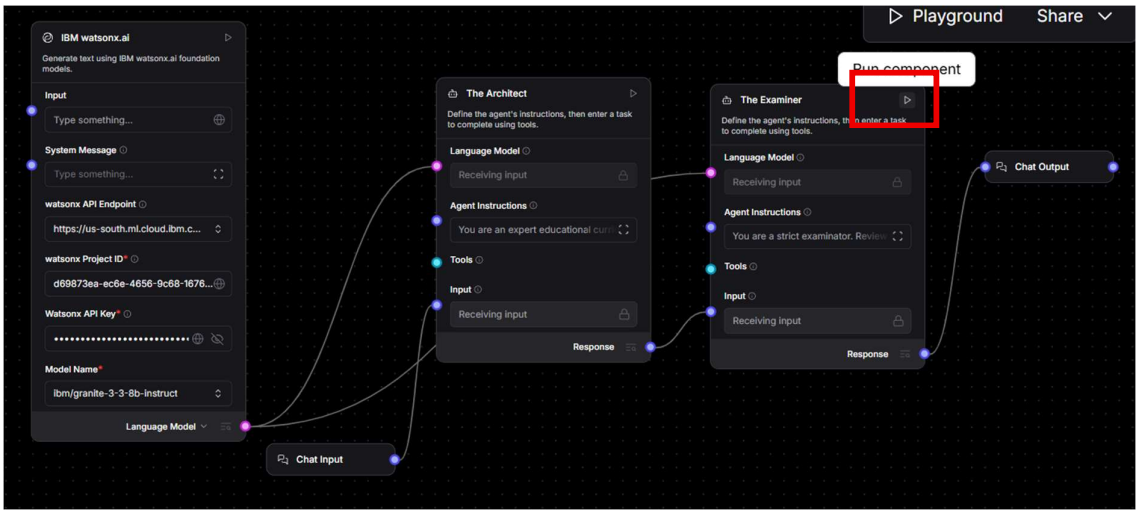
- Connect Agent2(The Examiner) response to chat output



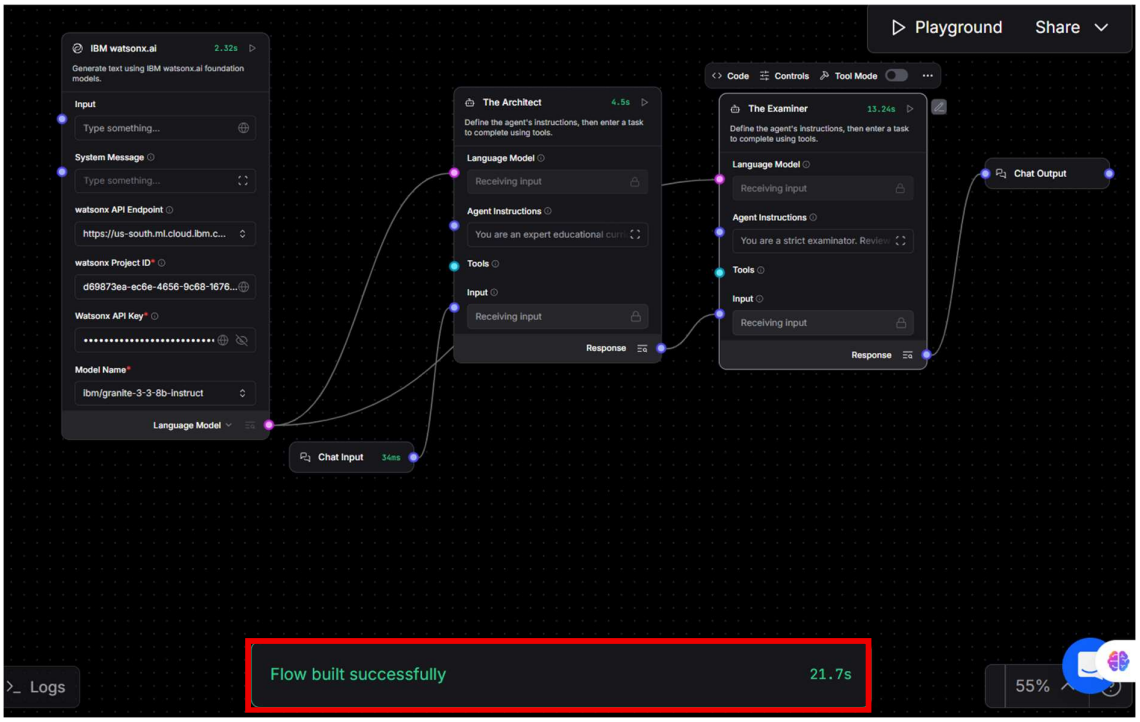
- Save the workflow (control + s) → The workflow will be saved



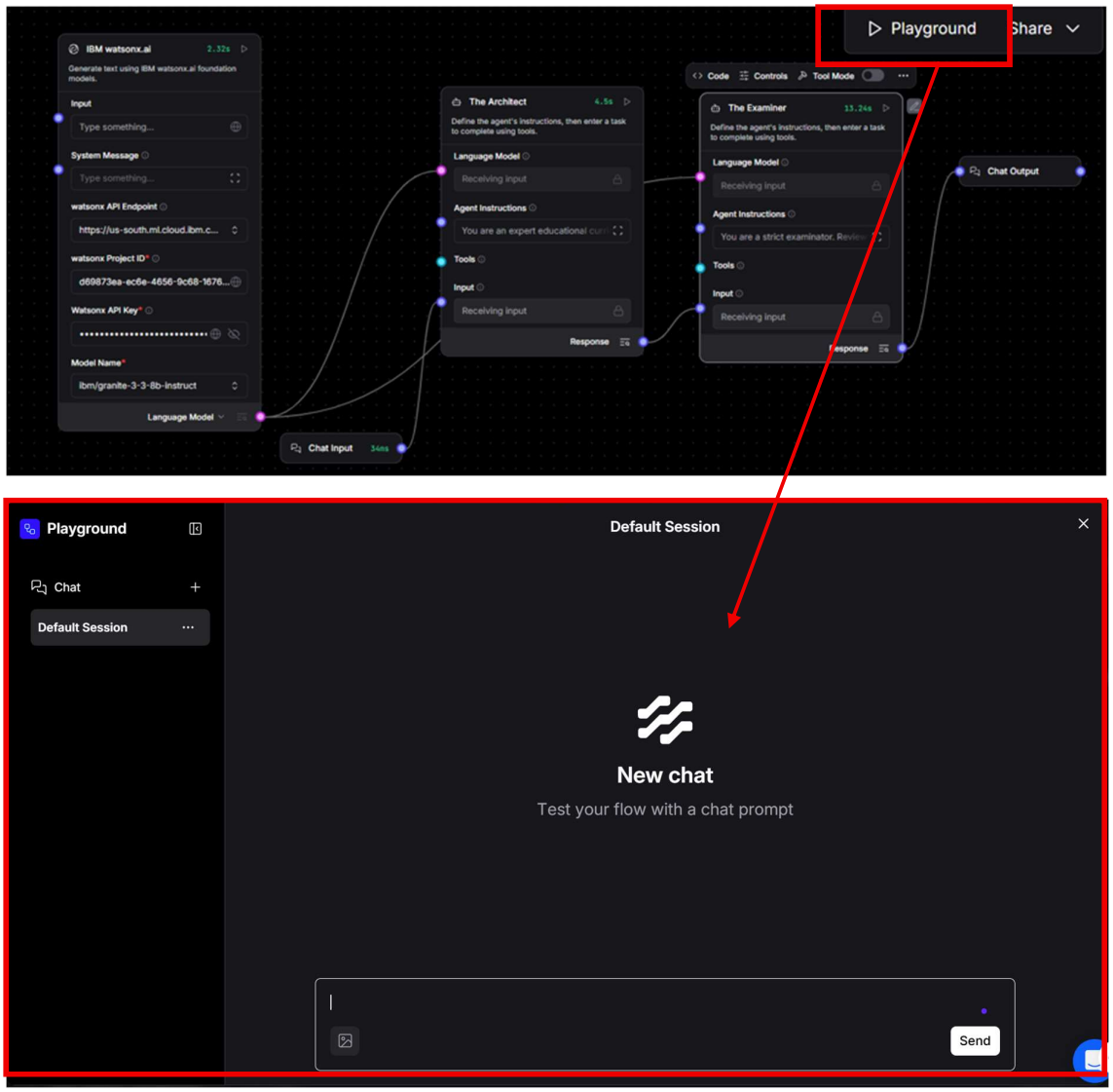
- Run Agent2(The Examiner) by clicking on Run symbol on the agent



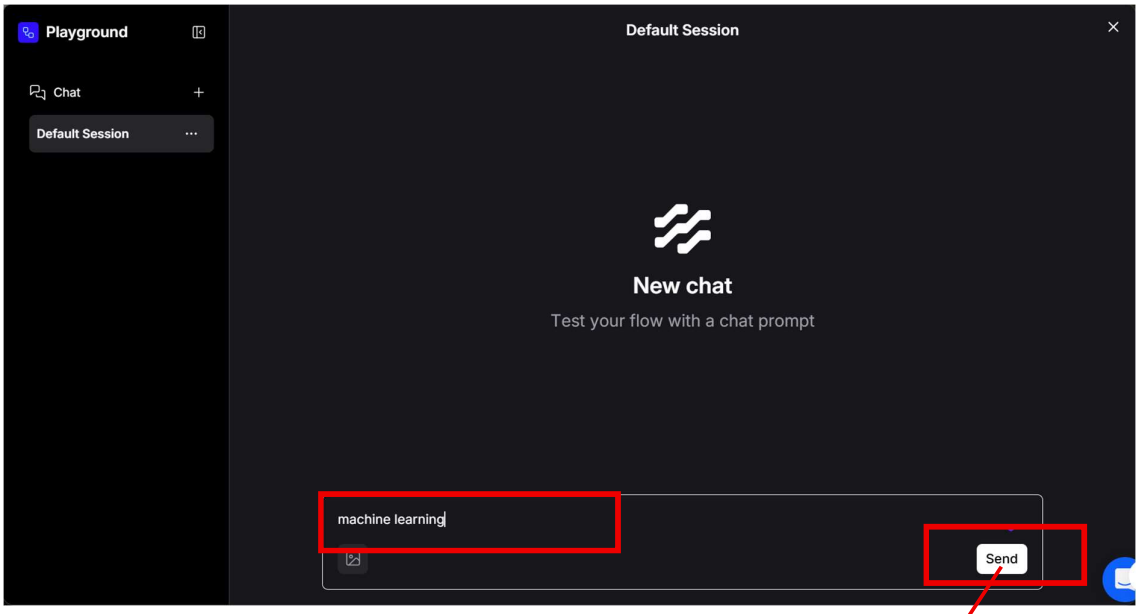
- The flow is built successfully now



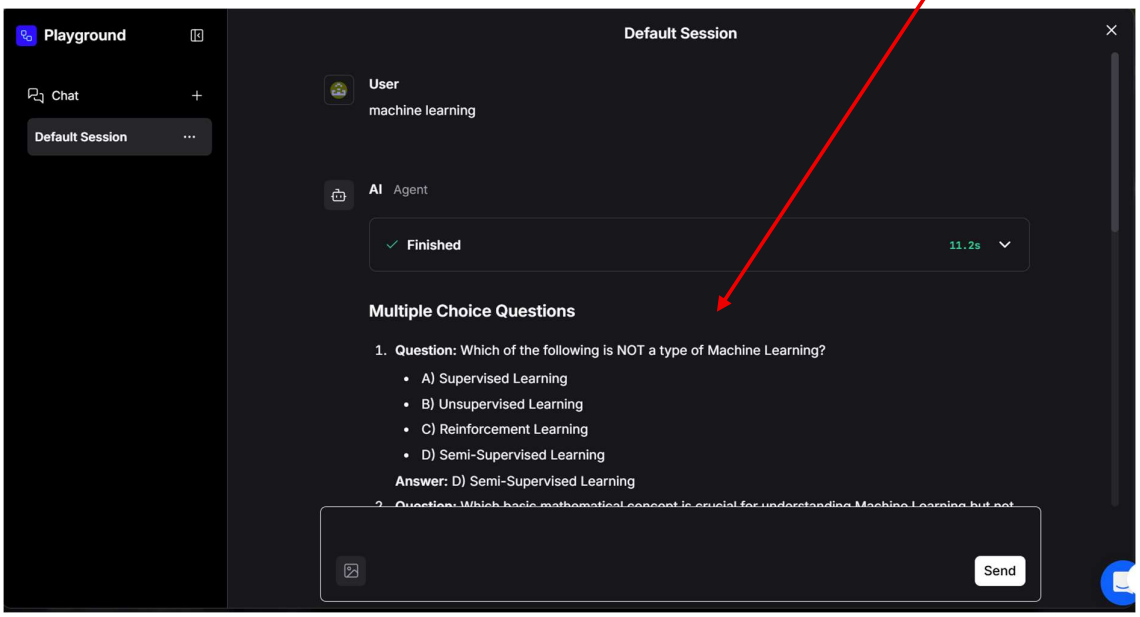
- Click on Playground button

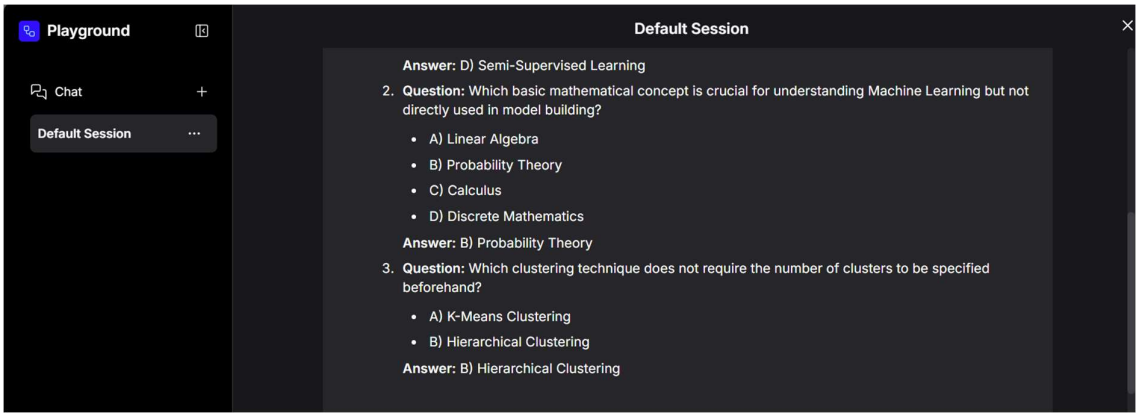


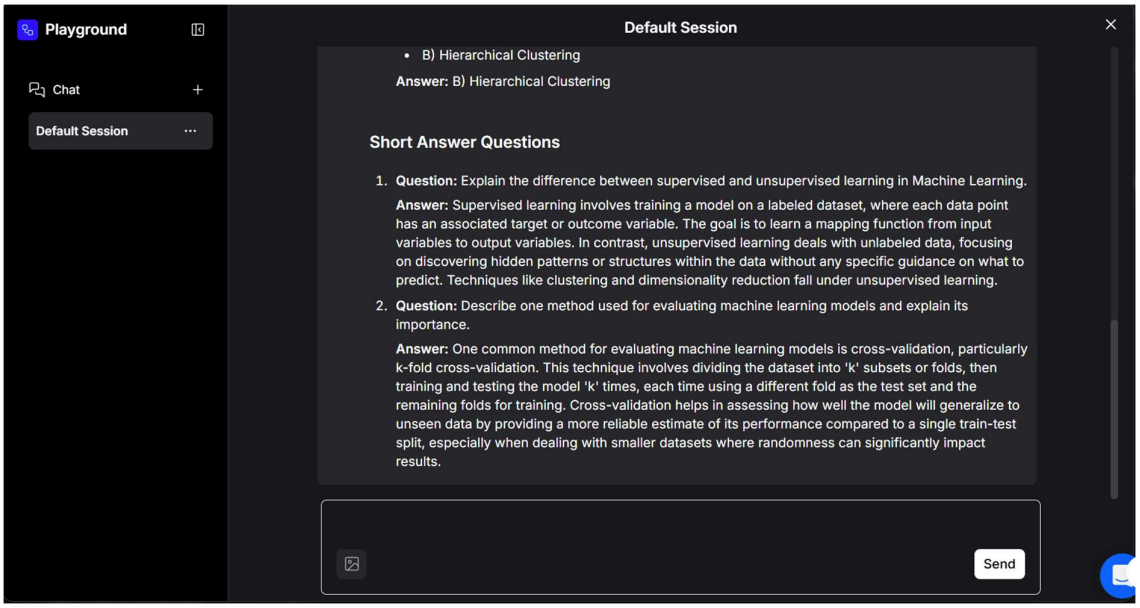
- Type machine learning in the text box → click on Send button



- See the output







- Congratulations agent working.